

Applying Humanistic Methods to Volumetric Capture

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Intro

Volumetric Capture (VolCap) uses depth-sensor cameras to create 3D representations of participants. Unlike traditional videos' fixed perspective, VolCap enables spatial agency: viewers can move around subjects, and the medium is flexible, allowing the same capture to be used for AR and VR experiences, or projection installations (Jones 2020). Our VolCap studio produced a campus historic site tour with a volumetric docent, positioning the subject within reconstructed spatial contexts. During the production of this project, we found ourselves questioning the best methods for conducting a capture for storytelling purposes. The current workflow for VolCap emphasizes how well the technology captures the physical appearance of a person, and tradeoffs are made based on the limitations of the technology and capture software. For example, restricting a performer's movement for the sake of "cleaner" data can alter how a person might move naturally. We propose to examine humanities professions that share similar objectives to help inform how the VolCap process can become more human-centered.

The VolCap Process and Its Limits

First, we need to thoroughly explain the VolCap process and highlight how the technical choices affect the final product, while emphasizing that the current focus is on technology rather than narrative. We opted to work with Depthkit as it provides a complete capture solution, has a lower barrier to entry, options for single or multisensor captures, and wide support (Depthkit Studio). The volumetric capture process can be split into three distinct subprocesses: the pre-capture, capture, and post-capture. It is imperative to understand how the individual parts of each of these subprocesses contribute to capturing the participant and require thoughtful decision-making.

Pre-capture

The pre-capture process involves determining the number of cameras required for the recording, their placement, lighting design, and initial system calibration. When planning a recording, it is important to understand the creative and narrative needs of the project. For example, is the subject seated, making a high-quality frontal view sufficient? Does the experience require a complete 360-degree view? Is a full-body capture necessary, or would a bust or mid-shot suffice? Answering these questions will help to define the required number of cameras and inform decisions about camera placement, lighting, and calibration. The final and most critical pre-capture step is camera calibration, to make sure that the image and depth data for each camera align correctly in the 3D space. Poor calibration can result in fragmented or blurry captures.

Capture

Communication with the participant becomes especially important during the capture. The director or technical lead should discuss what content is most valuable to record, including performance nuances, movement range, and whether the participant will remain stationary or move throughout the volume. Even small choices, such as how far the participant leans, turns, or extends their arms, can affect the quality and completeness of the volumetric recording and also affect the participant's expressiveness. In this way, the capture process remains a blend of technical precision and creative direction.

Post-capture

Volumetric data requires substantial processing and refinement during the post-capture phase. Adjustments can enhance depth map fusion and improve appearance by smoothing rough edges and repairing mesh geometry. However, excessive adjustments may cause distortions like warping. The team must determine when the capture has been refined sufficiently and decide what level of artifacting is acceptable, balancing creative goals and how the capture will be viewed.

In one case, smoothing reduced surface noise but unintentionally diminished the participant's ears. During review, they expressed feeling misrepresented. We returned to processing, adjusted smoothing parameters to preserve ear geometry while addressing other artifacts, and had the participant approve the revision, prioritizing their consent over technical optimization despite retained noise. This established a core principle: participant authority over their representation supersedes technical perfection. We now document these decisions alongside volumetric files, treating representation preferences as archival metadata. This iterative review process demonstrates how VolCap's detailed human data demands ethical protocols around consent, representation, and storage.

Methodological Transfers from the Humanities

We selected humanities professions sharing VolCap's goal, preserving individual and community stories, as methodological sources: oral history's collaborative storytelling (Yow 2014), anthropology's embodiment and mediated presence frameworks (Cirucci and Vandenberg 2023), and documentary filmmaking's narrative production ethics (Rabiger and Hurbis-Cherrier 2020). Through semi-structured interviews with practitioners and analysis of field-specific ethical guidelines, we are creating an integrated workflow. Validation will compare participant satisfaction, narrative coherence, and technical quality across projects using standard versus humanistic workflows. This approach centers human agency while treating technology as a recording method.

Proposed “Humanistic VolCap” Workflow

Drawing on humanities consultation and our docent project experience, we propose workflow modifications organized by production phase, grounding technical decisions in participant narrative authority.

Some initial observations:

Pre-Capture: Conduct narrative goal interviews establishing representation preferences and acceptable technical artifacts. Provide visual artifact examples so participants make informed consent decisions. Document these conversations as archival metadata.

Post-capture: Schedule participant review *during* processing, not only at completion. Record participant commentary on representation. Establish protocols where participant concerns override technical optimization.

Archiving & Dissemination: Store consent forms, representation preferences, and review commentary with volumetric files. Design exhibition contexts honoring agreed narrative framing. Establish participant-approved data retention and future use policies.

This process shifts towards storytelling, focusing on memory and narrative rather than just the surface appearance of the participant. This reframe positions VolCap as a tool for understanding and sharing people's stories, highlighting their humanity by exploring embodiment, identity, and presence through technology. The key question during the capture process changes from "How closely does this resemble the participant?" to "What narratives does this likeness convey?" We believe this work has larger implications and can be applied to all creative technology methods with humanistic subject matter.

Bibliography

Cirucci, Angela M. / Vandenberg, Amy (2023): "Volumetric Futures: Embodiment and Presence in 3D Capture", in: Media, Culture & Society 45.

Depthkit Studio (2025): "Depthkit Documentation" <https://docs.depthkit.tv/docs/depthkit-studio>.

Jones, Sarah (2020): "Volumetric Filmmaking and the Future of Immersive Media", in: Journal of Media Practice.

Nichols, Bill (2017): Introduction to Documentary. 3rd ed. Bloomington: Indiana University Press.

Rabiger, Michael / Hurbis-Cherrier, Mick (2020): Directing the Documentary. 7th ed. London: Routledge.

Yow, Valerie Raleigh (2014): Recording Oral History: A Guide for the Humanities and Social Sciences. 3rd ed. Lanham, MD: Rowman & Littlefield.

Interview questions:

Introduction

1. Could you please state and describe your area of expertise?
2. How long have you been working in this area? Were you formally trained?

Pre-Capture

1. What does “informed consent” look like in your field when the subject can’t fully preview the final product and how it will be used?
2. What training exists in your field around ethical representation that might not be obvious to someone outside the discipline?
3. How do you prepare participants before recording? How do you decide what information to provide about the process? What do you do to help familiarize them with the technology being used?
4. What conversations, if any, do you have with subjects about how they will be presented and what they would like to convey?

Capture

5. How do you handle the tension between technical quality and authentic representation?
6. How do you prioritize a participant’s self-representation over what you felt was a more urgent concern(e.g., deadlines, narratives, and technical complexity)? Have you made this decision, and what guided you?
7. What do you do when you notice a participant is self-conscious or sensationalizing?
8. How do you handle when a participant wants to revise or retract something during recording?

Post-Capture

9. What happens to the material you create?
 - a. Who controls the material?
 - i. Who decides how/if it is shared?
 - b. Who has access to it?
 - c. How is it being stored?
 - d. How long is it retained?
10. What do you do if a participant feels that they have been misrepresented after viewing the final product?

Interviewees

Digital Humanist

- ...

Oral Historians

- Jason Higgins

Anthropologists

- ...

Documentarians

- Charles Dye

IRB